

Derivatives of Functions

$$\frac{d}{dx} x^n = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} \tan x = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} e^x = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} \ln x = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} a^x = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} \arcsin x = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} \sin x = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} \arctan x = \underline{\hspace{2cm}}$$

$$\frac{d}{dx} \cos x = \underline{\hspace{2cm}}$$

Derivative Rules

Assume $f(x)$ and $g(x)$ are differentiable functions and c is a constant.

Constant Rule: $\frac{d}{dx} (cf(x)) = \underline{\hspace{2cm}}$

Sum Rule: $\frac{d}{dx} (f(x) + g(x)) = \underline{\hspace{2cm}}$

Product Rule: $\frac{d}{dx} (f(x)g(x)) = \underline{\hspace{2cm}}$

Quotient Rule: $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \underline{\hspace{2cm}}$

Chain Rule: $\frac{d}{dx} f(g(x)) = \underline{\hspace{2cm}}$

Derivative of an Inverse: $\frac{d}{dx} (f^{-1}(x)) = \underline{\hspace{2cm}}$